pgxnl 자연어 질의 테스트 결과 — BIRD mini-dev 82문항

측정일: 2026-10-07 (UTC 08:51~09:21) · 대상: pgxnl 자연어→SQL 질의 파이프라인 · 측정 run: card_games #384, california_schools #385

요약

항목	card_games	california_schools
문항 수	52	30
응답 시간 중앙값	13.0초	22.9초
응답 시간 평균	18.2초	28.2초
정답률 (공식 BIRD 채점)	41/52 (78.8%)	28/30 (93.3%)
동시 질문 수	1	1

1. 테스트에 사용한 질문

BIRD mini-dev 벤치마크(PostgreSQL판)의 영어 자연어 질문을 사용했습니다. 문항마다 데이터셋이 제공하는 보조 설명(evidence)이 함께 주어집니다. 전체 질문은 아래 부록: 82문항 기록에 있습니다.

스키마	문항 수	simple	moderate	challenging
card_games	52	13	33	6
california_schools	30	8	17	5
합계	**82**	21	50	11

2. 조회·연산한 데이터 규모

대상 테이블 크기

스키마	테이블	행 수
card_games	legalities	427,907

스키마	테이블	행 수
card_games	foreign_data	229,186
card_games	rulings	87,769
card_games	cards	56,822
card_games	set_translations	1,210
card_games	sets	551
california_schools	schools	17,686
california_schools	frpm	9,986
california_schools	satscores	2,269

질문 1건이 다룬 규모

지표	card_games	california_schools
참조 테이블 행 수 합 — 중앙값	58,032	19,955
참조 테이블 행 수 합 — 최대	572,498	29,941
결과 행 수 — 중앙값	1	1
결과 행 수 — 최대	29,936	7,806

- **참조 테이블 행 수 합**: 생성된 SQL이 사용한 테이블들의 행 수를 더한 값입니다. 실제로 읽은 행 수의 상한에 가깝습니다(인덱스로 일부만 읽을 수 있습니다).
- **결과 행 수**: 생성된 SQL을 실행해 돌려받은 행 수입니다. 대부분 집계·건수·비율처럼 1행이고, 목록형 질문은 수천~수만 행입니다.

3. 질문 1건당 응답 소요 시간

질문을 받은 뒤 SQL을 생성·검증·실행하고 답을 요약하기까지의 전체 시간입니다.

지표	card_games	california_schools
최소	7.8초	10.6초
중앙값	13.0초	22.9초

지표	card_games	california_schools
평균	18.2초	28.2초
90% 백분위	28.4초	46.5초
최대	85.3초	84.7초
전체 실행 시간	15분 48초 (52문항)	14분 7초 (30문항)

- 응답 시간의 대부분은 LLM 생성 시간입니다. LLM은 자체 서버 1대(NVIDIA L40S 1장, qwen3.8-27b, 단일 요청 기준 초당 약 19토큰)이고, 질문 1건에 LLM 호출이 여러 번 일어납니다(테이블 탐색 → SQL 생성 → 검증 → 답 요약). DB 실행 시간은 이에 비해 작습니다.
- 문항별 시간은 순차 처리된 문항의 완료 시각 차이로 계산했습니다.

4. 테스트 당시 동시 접속(동시 질문) 수

- **동시 질문 수: 1** — 평가는 동시성 1로, 질문을 한 번에 하나씩 순서대로 처리했습니다.
- **다른 사용자: 없음** — 측정 시간대의 감사 로그에는 평가 계정의 기록만 있습니다.
- **겹친 평가: 없음** — 두 run은 순서대로 실행됐습니다.

따라서 위 응답 시간은 **부하가 없는 단일 사용자 조건**의 값입니다. 동시 사용자가 늘면 같은 LLM 서버를 나눠 쓰므로 응답 시간이 늘어납니다. 동시 접속 성능은 별도로 측정하지 않았습니다.

부록: 82문항 기록

- **seq**: 평가 화면의 문항 번호 / **qid**: BIRD 원본 question_id
- **정답**: pgxnl 채점(중복 행 수까지 비교) / 공식 BIRD 채점(집합 비교). ✓ 맞음, ✗ 틀림
- **응답(초)**: 질문 1건 처리 시간 / **참조 행 수**: 사용한 테이블 행 수 합 / **결과 행**: 실행 결과 행 수

card_games (52문항, run #384)

seq	qid	난이도	질문	사용 테이블	참조 행 수	결과 행	응답(초)	정답 (pgxnl / 공식)
346	340	simple	Which are the cards that have incredibly powerful foils.	cards	56,822	25,061	15.9	✓ / ✓
347	341	simple	What are the borderless cards available without powerful foils?	cards	56,822	72	21.8	✓ / ✓
348	344	moderate	List all the mythic rarity print cards banned in gladiator format.	cards, legalities	484,729	5	18.3	✓ / ✓

seq	qid	난이도	질문	사용 테이블	참조 행 수	결과 행	응답(초)	정답 (pgxnl / 공식)
349	345	moderate	For artifact type of cards that do not have multiple faces on the same card, state its legalities status for vintage play format.	cards, legalities	484,729	3	12.4	✓ / ✓
350	346	moderate	List all the card id and artist with unknown power which are legal for commander play format.	cards, legalities	484,729	29,936	27.8	✓ / ✓
351	347	moderate	Find all cards illustrated by Stephen Daniel and describe the text of the ruling of these cards. State if these cards have missing or degraded properties and values.	cards, rulings	144,591	23	32.8	✓ / ✓
352	349	moderate	Name the card and artist with the most ruling information. Also state if the card is a promotional printing.	cards, rulings	144,591	1	12.8	✗ / ✗
353	352	moderate	Calculate the percentage of the cards availabe in Chinese Simplified.	foreign_data	229,186	1	20.1	✗ / ✗
354	356	simple	How many cards have infinite power?	cards	56,822	1	7.8	✓ / ✓
355	358	simple	What is the border color of card "Ancestor's Chosen"?	cards	56,822	1	8.2	✓ / ✓
356	366	simple	What is the rule of playing card "Benalish Knight"?	cards, legalities	484,729	32	10.5	✓ / ✓
357	368	simple	What is the percentage of borderless cards?	cards	56,822	1	8.9	✓ / ✓
358	371	challenging	What is the percentage of cards whose language is French among the Story Spotlight cards?	cards, foreign_data	286,008	1	31.1	✓ / ✓
359	377	simple	How many cards with original type of "Summon - Angel" have subtype other than "Angel"?	cards	56,822	1	9.8	✓ / ✓
360	379	simple	What are the cards belong to duel deck a? List the ID.	cards	56,822	804	15.2	✓ / ✓
361	383	simple	How many of the banned cards are white border?	cards, legalities	484,729	1	9.9	✓ / ✓
362	391	moderate	Among the Artifact card names, which are black color and comes with foreign languague translation?	cards, foreign_data	286,008	8	18.9	✓ / ✓
363	397	moderate	What is the mana cost of cards with a normal layout, a 2003 frame version, with a black border color, and available in paper and mtgo?	cards	56,822	11,446	20.7	✓ / ✓
364	402	moderate	What is the percentage of Story Spotlight cards that do not have a text box? List them by their ID.	cards	56,822	104	21.4	✗ / ✗
365	405	moderate	How many Brazilian Portuguese translated sets are inside the Commander block?	set_translations, sets	1,761	1	10.4	✓ / ✓
366	407	moderate	Lists all types of cards in German.	cards, foreign_data	286,008	408	26.5	✗ / ✓
367	408	moderate	How many unknown power cards contain info about the triggered ability	cards	56,822	1	9.3	✗ / ✗
368	409	moderate	Indicates the number of cards with pre-modern format, ruling text "This is a triggered mana ability." that do not have multiple faces.	cards, legalities, rulings	572,498	1	12.1	✓ / ✓

seq	qid	난이도	질문	사용 테이블	참조 행 수	결과 행	응답(초)	정답 (pgxnl / 공식)
369	412	moderate	What is the foreign name of the card in French of type Creature, normal layout and black border color, by artist Matthew D. Wilson?	cards, foreign_data	286,008	0	12.3	✗ / ✗
370	414	simple	What language is the set of 180 cards that belongs to the Ravnica block translated into?	set_translations, sets	1,761	10	20.5	✓ / ✓
371	415	challenging	What percentage of cards with format commander and legal status do not have a content warning?	cards, legalities	484,729	1	15.1	✓ / ✓
372	416	challenging	What percentage of cards without power are in French?	cards, foreign_data	286,008	1	38.0	✓ / ✓
373	422	simple	What is the language of the card with the multiverse number 149934?	foreign_data	229,186	1	11.8	✓ / ✓
374	424	simple	What proportion of cards do not have a text box with a normal layout?	cards	56,822	1	12.0	✗ / ✗
375	427	moderate	What languages are available in the set known as Archenemy on the magic card market and having the code ARC?	set_translations, sets	1,761	10	21.9	✓ / ✓
376	440	simple	Which foreign language used by "A Pedra Fellwar"?	foreign_data	229,186	1	11.1	✓ / ✓
377	459	moderate	Which card costs more converted mana, "Serra Angel" or "Shrine Keeper"?	cards	56,822	1	8.9	✓ / ✓
378	462	moderate	What's the Italian name of the set of cards with "Ancestor's Chosen" is in?	cards, set_translations	58,032	3	26.6	✗ / ✓
379	465	moderate	For the set of cards with "Ancestor's Chosen" in it, is there a Korean version of it?	cards, set_translations, sets	58,583	1	17.9	✗ / ✗
380	466	moderate	Among the cards in the set "Hauptset Zehnte Edition", how many of them are designed by Adam Rex?	cards, set_translations	58,032	1	11.0	✓ / ✓
381	468	moderate	What is the Simplified Chinese translation of the name of the set "Eighth Edition"?	set_translations, sets	1,761	1	9.4	✓ / ✓
382	469	moderate	Did the set of cards with "Angel of Mercy" appear on Magic: The Gathering Online?	cards, sets	57,373	15	10.8	✓ / ✓
383	472	moderate	Among the sets in the block "Ice Age", how many of them have an Italian translation?	set_translations, sets	1,761	1	10.4	✓ / ✓
384	473	moderate	Is the set of cards with Adarkar Valkyrie only available outside the United States?	cards, sets	57,373	5	11.5	✗ / ✗
385	474	moderate	Among the sets of cards that have an Italian translation, how many of them have a base set number of under 100?	set_translations, sets	1,761	1	22.2	✓ / ✓
386	477	challenging	Which of these artists have designed a card in the set Coldsnap, Jeremy Jarvis, Aaron Miller or Chippy?	cards, sets	57,373	2	11.1	✓ / ✓
387	479	moderate	Among the cards with converted mana cost higher than 5 in the set Coldsnap, how many of them have unknown power?	cards, sets	57,373	1	12.6	✓ / ✓
388	480	moderate	What is the Italian flavor text of the card "Ancestor's Chosen"?	cards, foreign_data	286,008	2	16.7	✓ / ✓
389	483	moderate	Please list the Italian text ruling of all the cards in the set Coldsnap.	cards, foreign_data, rulings, sets	374,328	68	85.3	✗ / ✗

seq	qid	난이도	질문	사용 테이블	참조 행 수	결과 행	응답(초)	정답 (pgxnl / 공식)
390	484	moderate	Please list the Italian names of the cards in the set Coldsnap with the highest converted mana cost.	cards, foreign_data, sets	286,559	12	28.4	✗ / ✗
391	486	moderate	What is the percentage of the cards with a converted mana cost of 7 in the set Coldsnap?	cards, sets	57,373	1	12.6	✓ / ✓
392	487	challenging	What is the percentage of incredibly powerful cards in the set Coldsnap?	cards, sets	57,373	1	13.2	✓ / ✓
393	518	moderate	Which of the play format has the highest number of banned status? Indicate the play format and the names of all the card meet the condition.	cards, legalities	484,729	546	30.0	✓ / ✓
394	522	moderate	Which cards are ranked 1st on EDHRec? List all of the cards name and its banned play format.	cards, legalities	484,729	66	11.4	✗ / ✓
395	528	challenging	List the names of all the cards in the set Hour of Devastation and find the formats in which these cards are legal.	cards, legalities, sets	485,280	1,890	17.0	✗ / ✓
396	529	moderate	Find and list the names of sets which doesn't have Japanese translation but have Korean translation.	set_translations, sets	1,761	0	10.1	✓ / ✓
397	530	moderate	List all the frame styles and cards Allen Williams worked on and find any banned cards if there are any.	cards, legalities	484,729	1,031	45.5	✗ / ✗

california_schools (30문항, run #385)

seq	qid	난이도	질문	사용 테이블	참조 행 수	결과 행	응답(초)	정답 (pgxnl / 공식)
438	5	simple	How many schools with an average score in Math greater than 400 in the SAT test are exclusively virtual?	satscores, schools	19,955	1	12.9	✓ / ✓
439	11	simple	Please list the codes of the schools with a total enrollment of over 500.	frpm	9,986	7,806	32.5	✓ / ✓
440	12	moderate	Among the schools with an SAT excellence rate of over 0.3, what is the highest eligible free rate for students aged 5-17?	frpm, satscores	12,255	1	20.5	✓ / ✓
441	17	simple	Rank schools by their average score in Writing where the score is greater than 499, showing their charter numbers.	satscores, schools	19,955	58	35.0	✓ / ✓
442	23	moderate	List the names of schools with more than 30 difference in enrollements between K-12 and ages 5-17? Please also give the full street adress of the schools.	frpm, schools	27,672	1,239	43.1	✓ / ✓
443	24	moderate	Give the names of the schools with the percent eligible for free meals in K-12 is more than 0.1 and test takers whose test score is greater than or equal to 1500?	frpm, satscores	12,255	1,167	22.5	✓ / ✓
444	25	moderate	Name schools in Riverside which the average of average math score for SAT is grater than 400, what is the funding type of these schools?	frpm, satscores, schools	29,941	6	36.9	✓ / ✓

seq	qid	난이도	질문	사용 테이블	참조 행 수	결과 행	응답(초)	정답 (pgxnl / 공식)
445	26	moderate	State the names and full communication address of high schools in Monterey which has more than 800 free or reduced price meals for ages 5-17?	frpm, schools	27,672	6	46.7	✓ / ✓
446	27	moderate	What is the average score in writing for the schools that were opened after 1991 or closed before 2000? List the school names along with the score. Also, list the communication number of the schools if there is any.	satscores, schools	19,955	813	46.5	✗ / ✗
447	28	challenging	Consider the average difference between K-12 enrollment and 5-17 enrollment of schools that are locally funded, list the names and DOC type of schools which has a difference above this average.	frpm, schools	27,672	57	57.0	✓ / ✓
448	31	moderate	What is the eligible free rate of the 10th and 11th schools with the highest enrolment for students in grades 1 through 12?	frpm	9,986	2	16.3	✓ / ✓
449	32	moderate	What is the eligible free or reduced price meal rate for the top 5 schools in grades 1-12 with the highest free or reduced price meal count of the schools with the ownership code 66?	frpm, schools	27,672	5	37.7	✓ / ✓
450	37	moderate	What is the complete address of the school with the lowest excellence rate? Indicate the Street, City, State and Zip.	satscores, schools	19,955	1	18.6	✓ / ✓
451	36	challenging	Under whose administration is the school with the highest number of students scoring 1500 or more on the SAT? Indicate their full names.	satscores, schools	19,955	1	42.2	✓ / ✓
452	39	simple	What is the average number of test takers from Fresno schools that opened between 1/1/1980 and 12/31/1980?	satscores, schools	19,955	1	13.6	✗ / ✗
453	40	moderate	What is the telephone number for the school with the lowest average score in reading in Fresno Unified?	satscores, schools	19,955	1	13.0	✓ / ✓
454	41	simple	List the names of virtual schools that are among the top 5 in their respective counties based on average reading scores.	satscores, schools	19,955	34	23.0	✓ / ✓
455	45	moderate	What is the average writing score of each of the schools managed by Ricci Ulrich? List the schools and the corresponding average writing scores.	satscores, schools	19,955	1	24.1	✓ / ✓
456	46	simple	Which state special schools have the highest number of enrollees from grades 1 through 12?	frpm, schools	27,672	1	13.1	✓ / ✓
457	47	moderate	What is the monthly average number of schools that opened in Alameda County under the jurisdiction of the Elementary School District in 1980?	schools	17,686	1	16.7	✓ / ✓
458	48	moderate	What is the ratio of merged Unified School District schools in Orange County to merged Elementary School District schools?	schools	17,686	1	17.4	✓ / ✓
459	50	simple	What is the postal street address for the school with the 7th highest Math average? Indicate the school's name.	satscores, schools	19,955	1	14.1	✓ / ✓

seq	qid	난이도	질문	사용 테이블	참조 행 수	결과 행	응답(초)	정답 (pgxnl / 공식)
460	62	challenging	What is the total number of non-chartered schools in the county of Los Angeles with a percent (%) of eligible free meals for grades 1 through 12 that is less than 0.18%?	frpm, schools	27,672	1	16.6	✓ / ✓
461	72	moderate	How many students from the ages of 5 to 17 are enrolled at the State Special School school in Fremont for the 2014-2015 academic year?	frpm, schools	27,672	2	30.9	✓ / ✓
462	77	moderate	Which schools served a grade span of Kindergarten to 9th grade in the county of Los Angeles and what is its Percent (%) Eligible FRPM (Ages 5-17)?	frpm, schools	27,672	2	31.8	✓ / ✓
463	79	moderate	Between San Diego and Santa Barbara, which county offers the most number of schools that does not offer physical building? Indicate the amount.	schools	17,686	1	22.7	✓ / ✓
464	82	simple	What is the grade span offered in the school with the highest longitude?	schools	17,686	1	10.6	✓ / ✓
465	83	challenging	Of the schools that offers a magnet program serving a grade span of Kindergarten to 8th grade, how many offers Multiple Provision Types? List the number of cities that offers a Kindergarten to 8th grade span and indicate how many schools are there serving such grade span for each city.	frpm, schools	27,672	1	84.7	✓ / ✓
466	85	moderate	What is the Percent (%) Eligible Free (K-12) in the school administered by an administrator whose first name is Alusine. List the district code of the school.	frpm, schools	27,672	1	24.9	✓ / ✓
467	87	challenging	What are the valid e-mail addresses of the administrator of the school located in the San Bernardino county, City of San Bernardino City Unified that opened between 1/1/2009 to 12/31/2010 whose school types are public Intermediate/Middle Schools and Unified Schools?	schools	17,686	1	21.9	✓ / ✓